

Fractions, Decimals & Percentages

Quick reference sheet · GCSE Maths, Number · keep handy for revision

1. Equivalences worth memorising

Fraction	Decimal	Percentage
$\frac{1}{2}$	0.5	50%
$\frac{1}{4}$	0.25	25%
$\frac{1}{5}$	0.2	20%
$\frac{1}{10}$	0.1	10%
$\frac{1}{8}$	0.125	12.5%
$\frac{1}{3}$	0.333...	33.3%

2. Converting between the three

Fraction → **Decimal**: divide top by bottom. e.g. $\frac{3}{4} = 3 \div 4 = 0.75$

Decimal → **Percentage**: $\times 100$ (move the point two places right). e.g. $0.75 \rightarrow 75\%$

Percentage → **Decimal**: $\div 100$ (move the point two places left). e.g. $45\% \rightarrow 0.45$

Decimal → **Fraction**: use place value, then simplify. e.g. $0.6 = \frac{6}{10} = \frac{3}{5}$

Fraction → **Percentage**: convert to a decimal first, then $\times 100$. e.g. $\frac{9}{20} = 0.45 = 45\%$

3. Ordering mixed values

Convert everything into the **same form** — decimals are usually quickest — then compare.

Example: order $\frac{7}{20}$, 30%, 0.28, $\frac{1}{3}$ → 0.35, 0.30, 0.28, 0.333... → **0.28, 30%, $\frac{1}{3}$, $\frac{7}{20}$**

4. Fraction / percentage of an amount

Fraction of an amount: $\div$ by the denominator, $\times$ by the numerator.

$\frac{3}{5}$ of 40 → $40 \div 5 = 8$, $8 \times 3 = \mathbf{24}$

Percentage of an amount (no calculator): find 10% ($\div 10$), then 1% ($\div 10$ again), and combine.

35% of £60 → 10% = £6, so 30% = £18, 5% = £3 → 30% + 5% = **£21**

5. Worked examples

Start	Working	Result
$\frac{3}{4}$	$3 \div 4 = 0.75$, $\times 100$	$0.75 \cdot 75\%$
0.6	$\frac{6}{10}$, simplify $\div 2$	$\frac{3}{5} \cdot 60\%$

45%

$\div 100$, place value

$0.45 \cdot 9/20$
